Exercise 6 DATE Fuctions

Table 1: Employees

emp-id	name	hire-date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Question 1

Add 6 months to each employee's hire date using DATEADD().

Expected columns: emp-id, name, hire-date, hire-plus-6-months

```
SELECT emp-id, name, hire-date,
        DATEADD(month, 6, hire-date) AS hire-plus-6-months
FROM Employees;
```

emp-id	name	hire-date	hire-plus-6-months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

student_id	name	dob
101	Maya	2005-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

Question 2

Use DATEDIFF() to find age in days from dob to today.
Expected column: student_id, name, age_in_days

```
SELECT student_id, name,
        DATEDIFF(day, dob, CURRNT_DATE) AS age_in_day
FROM Students;
```

student_id	name	age_in_day
101	Maya	7560
102	Ethan	7512
103	Sienna	7343

Table 3: Events

event_id	event_name	event_date
1	Seminar	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

Question 3

Find how many days are left until each event using DATEDIFF().
Expected columns: event_id, event_name, days_remaining

```
SELECT event_id, event_name,
        DATEDIFF(day, CURRENT_DATE, event_date) AS days_remaining
FROM Events;
```

event_id	event_name	days_remaining
1	Seminar	-676
2	Workshop	-233
3	Hackathon	-457

Table 4 : Invoices

invoice_id	issue_date	due_date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

Question 4

Calculate the number of days between issue_date and due_date.

Expected columns : invoice_id, issue_date, due_date, days_between

```
SELECT invoice_id, issue_date, due_date,
        DATEDIFF(day, issue_date, due_date) AS days_between
FROM Invoices;
```

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5: Courses

course_id	name	start_date
201	SQL Basics	2025-05-01
202	Python	2025-06-01

Question 5

Formate start_date as 'Month YYYY' using TO_CHAR().
Expected columns: course_id, name, formatted_date

```
SELECT course_id, name,
        TO_CHAR(start_date, 'MMMM yyyy') AS formatted_date
FROM Courses;
```

course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6: Memberships

member_id	start_year	start_month	start_day
1	2023	5	10
2	2022	11	25

Question 6

Create full date from parts using DATE_FROM_PART().
Expected columns: member_id, full_start_date

```
SELECT member_id,
        DATE_FROM_PART( start_year, start_month, start_day) AS
        full_start_date
FROM Memberships;
```

member_id	full_start_date
1	2023-05-10
2	2022-11-25

Table 7: Subscriptions

sub_id	plan	renewal_date
11	Basic	2025-01-01
12	Premium	2025-03-15

Question 7

Extend each renewal_date by 1 year using DATEADD().
Expected columns : sub_id, plan, extended_renewal_date

```
SELECT sub_id, plan,
        DATEADD(year, 1, renewal_date) AS extended_renewal_date
FROM Subscriptions;
```

sub_id	Plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Table 8: Orders

order_id	order_date
1001	2025-04-15
1002	2025-04-10

Question 8

Show current date and difference from order_date. Use CURRENT_DATE and DATEDIFF().

Expected columns: order_id, order_date, today_date, days_since_order

```
SELECT order_id, order_date,
        CURRENT_DATE AS today_date,
        DATEDIFF(CURRENT_DATE, order_date) AS
        days_since_order
FROM Orders;
```

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-23	373
1002	2025-04-10	2026-04-23	378

```
SELECT order_id, order_date,
        CURRENT_DATE AS today_date,
        DATEDIFF(CURRENT_DATE, TRY_CAST( order_date
        AS DATE)) AS days_since_order
FROM Orders;
```

Table 9: Trainings

training_id	topic	training_date
1	Safety	2025-01-10
2	Compliance	2025-02-20

Question 9

Extract the year from training_date using DATE_PART() or EXTRACT().

Expected columns: training_id, topic, training_year

```
SELECT training_id, topic,
        EXTRACT(YEAR FROM training_date) AS training_year
FROM Trainings;
```

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

Table 10: Blog_Posts

post_id	title	published_on
1	SQL Tips	2025-04-01 10:15:00
2	Data Cleaning	2025-04-12 16:45:00

Question 10

Extract hour and minute from published_on

Expected columns: post_id, title, hour_published, minute_published

```
SELECT post_id, title,
        EXTRACT(hour FROM published_on) AS hour_published,
        EXTRACT(minute FROM published_on) AS
        minute_published
FROM Blog_posts;
```

post_id	title	hour_published	minute_published
1	SQL Tips	10	15
2	Data Cleaning	16	45

Table 11 : Drivers

driver_id	license_expiry
301	2025-08-10
302	2023-12-31

Question 11

Calculate days left until licence expiry using DATEDIFF() and today's date.

Expected column : driver_id, licence_expiry, days_left

```
SELECT driver_id, licence_expiry,
        DATEDIFF(day, CURRENT_DATE, TRY_CAST(licence_expiry
        AS DATE)) AS days_left
FROM Drivers;
```

driver_id	licence_expiry	days_left
301	2025-08-10	-256
302	2023-12-31	-844

Table 12: Messages

message_id	sent_timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

Question 12

Display the current timestamp and calculate seconds since the message was sent

Expected columns: message_id, sent_timestamp, current_timestamp, seconds_since_sent

```
SELECT message_id,
        TRY_TO_TIMESTAMP(sent_timestamp) AS clean_timestamp,
        DATEDIFF(second, TRY_TO_TIMESTAMP(sent_timestamp),
        CURRENT_TIMESTAMP) AS seconds_since_sent
FROM messages;
```

message_id	clean_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	31899612
2	2025-04-18 23:59:59	31933978

Table 13: Returns

return_id	return_date
901	2025-04-05
902	2025-04-01

Question 13

Add 15 days to return-date using DATEADD() to show restock-date.

Expected columns: return-id, return-date, restock-date

```
SELECT return-id, return_date,
        DATE_ADD(return_date, 15) AS restock_date
FROM Returns;
```

return-id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14: Assignments

assign-id	assigned-on
1	2025-03-01
2	2025-03-05

Question 14

Convert assigned-on to date using TO-DATE() (if it's stored as string).

Expected columns: assign-id, assigned_on_date

```
SELECT assign-id,
        TO_DATE(assigned-on) AS assigned-on-date
FROM Assignments;
```

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

Table 15: Meetings

Meeting_id	Scheduled_time
1	2025-04-19 14:00:00
2	2025-04-19 09:30:00

Question 15

Convert scheduled_time to formatted string like 'April 19, 2025 at 2:00 PM' using TO_CHAR().

Expected columns: meeting_id, formatted_meeting_time

```
SELECT meeting_id,
        CONCAT(DATE_FORMAT(CAST(scheduled_time
        AS TIMESTAMP), 'MMMM dd, y')),
        'at', DATE_FORMAT(CAST(scheduled_time
        AS TIMESTAMP), 'h:mm a')) AS
        formatted_meeting_time
FROM Meetings,
```

meeting_id	formatted_meeting_time
1	April 19, 2025 at 2:00 PM
2	April 19, 2025 at 9:30 AM